

Diabetes Risk Prediction Using Machine Learning

This report summarises an end-to-end computational health analytics pipeline for diabetes risk prediction. The workflow includes: - Data preprocessing - Feature engineering - Machine learning modelling - Clinical validation - Explainable AI - Treatment effect analysis

Machine Learning Model Performance

Model	Accuracy	Precision	Recall	F1	ROC_AUC
Logistic_regression	0.7077922077922078	0.6	0.5	0.5454545454545454	0.8148148148148148
Random_forest	0.7467532467532467	0.6595744680851063	0.5740740740740741	0.6138613861386139	0.8214814814814814
Gradient_boost	0.7532467532467533	0.66	0.6111111111111112	0.6346153846153846	0.8225925925925925

Treatment Effect Analysis

Metric	Value
Relative Risk	1.244087481852667
Risk Difference	0.0786289714268472
Odds Ratio	1.407330702199992
Chi-square p-value	0.0368388224622952

Important Predictive Variables

The predictive modelling analysis identified clinical variables associated with diabetes risk. Important variables included:

Treatment : -0.227

Age : 0.035

BMI : 0.103

Glucose : 0.039

BloodPressure : -0.003

Model Validation Figures



